

GESIS Fall Seminar in Computational Social Science 2026

“Introduction to Machine Learning for Text Analysis with Python”

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Time: 09:45-16:45 (CEST/UTC+2)
Location: GESIS Mannheim

About the Lecturers

Sjoerd Stolwijk is a Senior Researcher in the School of Governance at Utrecht University. He is part of the RadiOnce consortium led by Prof. Dr. Barbara Vis, where he investigates the influence of uncertainty on political decision-making. He is also part of the EU COST OPINION network, where he works on recommendations for the best automatic measures for opinions in text. Previously, he was a Postdoc in Computational Communication at the University of Amsterdam and part of the TWON consortium led by Prof. Dr. Damian Trilling, where he investigated how to use automatic methods to measure deliberative discourse online. He is currently working on annotation biases of both BERT and generative AI methods. He completed his PhD in political communication at the University of Amsterdam in 2017.

Rupert Kiddle is a Research Associate at the Department of Communication Science at Vrije Universiteit Amsterdam. He is part of the WHAT-IF (ERC Horizon) project, led by Prof. Dr. Wouter van Atteveldt, which seeks to develop a digital twin of the political information environment. His work there involves using generative AI together with agent-based modeling to simulate immersive platform environments for supporting more ecologically valid participant-centered studies. Previously, as part of his doctoral work, he was part of the NEWSFLOWS (ERC Horizon) project, led by Prof. Dr. Damian Trilling, which used language modelling in order to understand (through modeling data donations) and influence (through recommender system designs) so-called “feedback loops” in individual-level media (news) selection behavior.

Course Description

The course will provide insights into the concepts, challenges, and opportunities associated with textual data that is so large that traditional research methods (like manual content analysis) can no longer be applied, and traditional inferential statistics start to lose their meaning. In this course, you will be introduced to strategies and techniques for capturing and analyzing textual data in social science and communication contexts using Python. The course offers hands-on instructions regarding the several stages of computer-aided content analysis. More precisely, you will be familiarized with pre-processing methods, analysis strategies, and the presentation of findings. The focus will be in particular on machine learning techniques to analyze quantitative textual data, amongst which both deductive (e.g., supervised machine learning) and inductive (e.g., unsupervised machine learning) approaches will be discussed, with special emphasis on the use of transformer models (e.g., BERT).

This is a beginner’s course. While we do work with state-of-the-art methods (e.g., BERTopic, fine-tuning a BERT-based classifier), we will present them in a simplified manner for those new to machine learning for textual analysis. While extensive programming experience is not required for this course, **you will need to have a basic grasp of Python** (see *Course Prerequisites*). If you already have more experience with language modelling (e.g., working with embeddings) or are looking to work with generative large language models (e.g., ChatGPT), you should consider taking the more advanced course “[From Embeddings to LLMs: Advanced Text Analysis in Python](#)” (21-25 September 2026).

Organizational Structure of the Course

In the mornings, we will have lectures in which we will explain the topic of the day from both a theoretical-conceptual point of view as well as from a practical point of view (i.e., walking you through code examples). This will sometimes include small in-class exercises.

In the afternoon, you will work on larger exercises in which you implement the techniques we covered. We provide example datasets, but it is also possible (and strongly encouraged) to try to apply the techniques to your own datasets (see *Course Prerequisites*). You can either opt to work on your own or try to solve problems together with one or multiple classmates. Lecturers will provide guidance and feedback throughout.

The first three days (Mon-Wed) of the course focus on discussing and applying new concepts (see *Day-to-day Schedule* below). The final two days (Thu-Fri) of the course are focused on application and will have more time for you to work on your own dataset or research project. There will also be optional mini-lectures on more advanced topics (e.g., evaluation best practices, navigating HuggingFace, explainable AI) with room for discussion.

During the course, we provide all code as notebooks with extensive commentary to explain what action is being performed at each step and why. These are intended to aid understanding for those who are relatively new to Python. In addition, the lecturers will provide an introduction to using AI for coding, which is not essential for the course, but can be a very useful tool for learning a new programming language if used responsibly.

Keywords

Language Modelling, Transformers, Supervised Machine Learning, Unsupervised Machine Learning

Target Group

You will find the course useful if:

- You are a social scientist who aims to describe, explain, or predict the content of large-scale textual data using computational techniques.
- You may have some awareness of or experience with language modelling methods, but if you do, you have not worked with them in-depth.
- You desire hands-on experience with how to implement such methods for unsupervised (e.g., topic modeling) or supervised machine learning (e.g., classification) research tasks.

Note that non-textual data, such as images, videos, or multi-modal data, are not the focus of this course. The techniques we cover are partly generalizable to such types of data, but the course is not tailored towards them. If you want to work with videos, images, and/or audio, you might be interested in the course "[Computer Vision for Image and Video Data Analysis with Python](#)" (14-18 September 2026).

Course and Learning Objectives

By the end of the course, you will:

- Be able to identify and situate research methods from computer science and computational linguistics, which can be used for research in the domain of the social sciences.
- Understand the principles of supervised and unsupervised machine learning.
- Be able to explain the principles of these methods and describe the value of these methods.
- Know how to use relevant Python modules for questions relevant in the domain of the social sciences.
- Be able to independently analyze quantitative textual data using state-of-the-art machine learning techniques.

Course Prerequisites

- Some knowledge of basic statistics (linear and logistic regression).
- Some basic experience with Python and/or other programming/statistical languages: While extensive programming experience is not required for this course, **you will need to have a basic grasp of Python to be able to follow the course**. We have prepared the following two introductory notebooks that we ask you to complete beforehand to check/refresh your knowledge (see setup guide below). These notebooks contain links to the relevant sections of the main coursebook (<https://cssbook.net/>).
 - [Introduction to Python](#): Introduction, data types, control structures, and dealing with errors.
 - [Loading and Exploring Data](#): Walkthrough of loading and exploring datasets used in the course.

If you are completely new to Python, we recommend you register for the introductory online workshop “[Introduction to Python](#)” (25-27 August 2026) in addition.

- Finally, you are **strongly encouraged to bring your own dataset** to make use of during the course. This can either be a:
 1. labeled dataset:
containing at least a column with text entries and one with labels (e.g., “positive”, “negative”), which you can use to train a supervised model, to predict some quantity or category.
 2. unlabeled dataset:
containing only text fields (on which you can do some unsupervised machine learning, such as topic modelling, to explore and describe the data).

You may bring a dataset in any format (e.g., CSV or JSON), though, as the focus of the course will not be on data wrangling, please ensure that you know how to load and explore your dataset (for example, converting it into a data frame; see [Loading and Exploring Data](#)). If you are unable to bring a dataset for any reason, the lecturers will provide a small selection to choose from.

Software and Hardware Requirements

Prior to the course, please consult this [setup guide](#). This will guide you through setting up a Python programming environment and cloning (copying) the course materials to your laptop. **It is important that you complete this setup before attending the course**, and potentially well in advance in the case of ‘managed laptops’, since your employer may need to be involved in the setup (e.g., if additional permissions are needed to install software).

We will be working with computationally demanding models and methods. We therefore ask that you determine whether your own laptop will be capable of supporting these:

- **Windows/Linux**: i5 or Ryzen 5 CPU, > 16GB RAM, with a dedicated NVIDIA GPU with > 4GB VRAM.
- **Mac**: any ‘M’ generation laptop (M1-M4) with at least 8GB of unified memory (better: 16GB+).

Detailed instructions for evaluating whether your laptop meets the above demands can be found in the [setup guide](#). Please reach out to the lecturers if you need assistance evaluating your device against these minimums.

If your device does not meet these minimums, we recommend that you sign up for *Google Colab* (see the *second part of the setup guide*). This is a cloud-based alternative to using your own laptop to perform computations. GPU time on *Google Colab* is charged, but relatively inexpensive. You should budget a maximum of €20 for GPU credits (e.g. in the form of a monthly *Pro* plan subscription; the current price of the *Pro* plan (March 2026) is €11.01 per month), in case the free quota does not suffice for the course. Using *Google Colab* also requires a Google account. Alternatively, you may wish to make use of a compute environment provided by your institution (e.g., SURF in the Netherlands). You are free to do so, but note that the lecturers may be limited in their capacity to resolve any issues that arise within such environments.

Day-to-day Schedule and Literature

Day 1: Introduction to Text-as-Data: From Dictionaries to Contextual Embeddings

- Introduction and overview of the course
- Inductive and deductive approaches to automated content analysis
- Situating methods: rule-based, supervised machine learning (SML), unsupervised machine learning (UML)
- Hands on with text-as-data: from dictionaries to bag-of-words, to contextual embeddings
- The workflow: preprocessing, vectorization, dimensional reduction and clustering, modelling, evaluation
- Housekeeping: accessing the course materials, managing your code environment, how to use AI tools

For day one, the morning will focus on a minimal-code introduction to the most important ideas and computational-linguistic methods covered by the course, including a demonstration of different representation methods (i.e., turning text into numbers) and a discussion of their strengths and limitations. In the afternoon, there will be the option to work on some basic text preprocessing exercises and/or ask questions about your own (BYO) datasets and potential research questions to investigate during this course.

Literature:

van Atteveldt, W., Trilling, D., & Arcila Calderon, C. (2022). *Computational Analysis of Communication: A Practical Introduction to the Analysis of Texts, Networks, and Images with Code Examples in Python and R*. Hoboken, NJ: Wiley. Chapters **1-7** (optional, for Python basics), **9-11** (for introduction to text-as-data). <https://cssbook.net/>

Boumans, J. W. & Trilling, D. (2016). Taking Stock of the Toolkit. An Overview of Relevant Automated Content Analysis Approaches and Techniques for Digital Journalism Scholars. *Digital Journalism*, 4(1), 8-23. [doi:10.1080/21670811.2015.1096598](https://doi.org/10.1080/21670811.2015.1096598)

Kroon, A., Welbers, K., Trilling, D., & van Atteveldt, W. (2024). Advancing Automated Content Analysis for a New Era of Media Effects Research: The Key Role of Transfer Learning. *Communication Methods and Measures*, 18(2), 142-162. [doi:10.1080/19312458.2023.2261372](https://doi.org/10.1080/19312458.2023.2261372)

Day 2: Unsupervised Machine Learning with BERTopic

- Principles of unsupervised machine learning
- Introduction to tokenization and vectorization
- Introduction to dimensional reduction and clustering
- Hands-on topic modelling with transformers: a simple BERTopic pipeline
- (Optional) Advanced BERTopic: semi-supervised, temporal modelling, GLLMs

For day two, we will spend the morning introducing the concept and empirical use-cases of unsupervised machine learning (UML) and introducing key steps in a typical unsupervised pipeline: tokenization, vectorization, dimensional reduction, and clustering. In the afternoon, we will use BERTopic to produce and explore a topic model.

Literature:

van Atteveldt, W., Trilling, D., & Arcila Calderon, C. (2022). *Computational Analysis of Communication: A Practical Introduction to the Analysis of Texts, Networks, and Images with Code Examples in Python and R*. Hoboken, NJ: Wiley. Chapters **7.3** and **11.5**. <https://cssbook.net/>

Grootendorst, M. (2022). BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure. arXiv. <https://arxiv.org/abs/2203.05794>

Day 3: Supervised Machine Learning with BERT

- Principles of supervised machine learning
- Starting small: a dictionary-based example
- Simple bag-of-words model: Naive Bayes
- Introduction to evaluation: performance metrics
- What is fine-tuning, and what does it achieve?
- Hands-on fine-tuning and evaluating a simple BERT-based classifier

For day three, the morning will be spent introducing the concept and empirical use cases of supervised machine learning (SML). From there, we will start small: with a dictionary-based approach and move on to a Naive Bayes approach with a bag-of-words. We will round out the morning by discussing how to evaluate SML classifier performance. In the afternoon, we will turn to the state of the art: fine-tuning a BERT-based classifier to see if we can further improve this performance.

Literature:

van Atteveldt, W., Trilling, D., & Arcila Calderon, C. (2022). *Computational Analysis of Communication: A Practical Introduction to the Analysis of Texts, Networks, and Images with Code Examples in Python and R*. Hoboken, NJ: Wiley. Chapters 8-10. <https://cssbook.net/>

Siino, M., Tinnirello, I., & La Cascia, M. (2024). Is Text Preprocessing Still Worth the Time? A Comparative Survey on the Influence of Popular Preprocessing Methods on Transformers and Traditional Classifiers. *Information Systems*, 121(C). [doi:10.1016/j.is.2023.102342](https://doi.org/10.1016/j.is.2023.102342)

Day 4: Best Practices for Supervised Machine Learning + Project Day 1

- Improving your SML workflow: best practices, test-train splitting, and overfitting
- Evaluation metrics explained: F1, precision, recall
- Advanced SML: cross-validation, hyperparameters, and grid search
- Practicalities: sample size and model performance

For day four, we will spend the morning starting our applied SML workflow. In the afternoon, you can work on implementing your preferred method (UML/SML) for your own use case, using your own dataset or a dataset provided by the lecturers. Lecturers will help where needed.

Literature:

Timoneda, J. C., & Vallejo Vera, S. V. (2025). BERT, RoBERTa, or DeBERTa? Comparing Performance across Transformers Models in Political Science Text. *The Journal of Politics*, 87(1), 347-364. [doi:10.1086/730737](https://doi.org/10.1086/730737)

Day 5: Navigating the Model Ecosystem, Critical Evaluation, and Generative AI + Project Day 2

- Navigating the model ecosystem (an introduction to HuggingFace)
- Model bias and the 'best' model for your task and corpus.
- Adaptation: continuous pre-training and (parameter-efficient) fine-tuning

Day 5 is for broadening our options, exploring possible additional techniques and platforms to find models and data. In the afternoon, you may optionally present your research projects to get feedback from the lecturers and audience.

Literature:

Balluff, P., Ho, J. C. T., Gruber, J. B., Palicki, S., Palmer, A., Rossi, L., ... & Chan, C. H. (2026). Newer, Larger, Better? A Critique of the Unreflective LLM Adoption in Communication Research. *Political Communication*, 1-10. [doi:10.1080/10584609.2026.2618486](https://doi.org/10.1080/10584609.2026.2618486)

Stolwijk, S. B., Boukes, M., Yeung, W. N., Liao, Y., Munker, S., Kroon, A. C., & Trilling, D. (2025). Can We Use Automated Approaches to Measure the Quality of Online Political Discussion? How to (Not) Measure Interactivity, Diversity, Rationality, and Incivility in Online Comments to the News. *Communication Methods and Measures*, 20(1), 1-25. [doi:10.1080/19312458.2025.2553300](https://doi.org/10.1080/19312458.2025.2553300)

Additional Recommended Literature

Brandt, P. T., Alsarra, S., D'Orazio, V., Heintze, D., Khan, L., Meher, S., ... Sianan, M. (2025). Extractive versus Generative Language Models for Political Conflict Text Classification. *Political Analysis*, 1–29. [doi:10.1017/pan.2025.10027](https://doi.org/10.1017/pan.2025.10027)

Licht, H., Sarkar, R., Wu, P. Y., Goel, P., Stoehr, N., Ash, E., & Hoyle, A. M. (2025). Measuring Scalar Constructs in Social Science with LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 32132–32159. [doi:10.18653/v1/2025.emnlp-main.1635](https://doi.org/10.18653/v1/2025.emnlp-main.1635)